

Coefficient of Variation

We noted earlier that the standard deviation is more easily interpreted than variance because standard deviation uses the same units of measurement as the observations. We may sometimes find it difficult to interpret what standard deviation means in terms of the relative degree of variability of different sets of data, however, either because the datasets have markedly different means or because the datasets have different units of measurement. In this section, we explain a measure of relative dispersion, the coefficient of variation that can be useful in such situations. **Relative dispersion** is the amount of dispersion relative to a reference value or benchmark.

The coefficient of variation is helpful in such situations as that just described (i.e., datasets with markedly different means or different units of measurement).

Coefficient of Variation Formula. The **coefficient of variation** (CV) is the ratio of the standard deviation of a set of observations to their mean value:

$$CV = \frac{s}{\bar{X}}, \quad (7)$$

where s is the sample standard deviation and $\bar{X}$ is the sample mean.

When the observations are returns, for example, the CV measures the amount of risk (standard deviation) per unit of reward (mean return). An issue that may arise, especially when dealing with returns, is that if $\bar{X}$ is negative, the statistic is meaningless.

The CV may be stated as a multiple (e.g., two times) or as a percentage (e.g., 200 percent). Expressing the magnitude of variation among observations relative to their average size, the CV permits direct comparisons of dispersion across different datasets. Reflecting the correction for scale, the CV is a scale-free measure (i.e., it has no units of measurement).

We illustrate the usefulness of CV for comparing datasets with markedly different standard deviations using two hypothetical samples of companies in Example 5.

EXAMPLE 5

Coefficient of Variation of Returns on Assets

Suppose an analyst collects the return on assets (ROA), in percentage terms, for 10 companies for each of two industries:

Exhibit 12: Returns on Assets for Two Industries

Company	Industry A	Industry B
1	-5	-10
2	-3	-9
3	-1	-7
4	2	-3
5	4	1
6	6	3
7	7	5
8	9	18
9	10	20
10	11	22

These data can be represented graphically as shown in Exhibit 13: